

NIKITHA THODUGULI

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EDUCATION

Massachusetts Institute of Technology (June 2027) | Cambridge, MA | GPA: 5.0/5.0

Major: Computer Science & Molecular Biology; Minors: Neuroscience, Political Science

LEADERSHIP

President, MIT Biotech Group (Fall 2026 – present)

Oversee six initiatives and 100+ members, lead annual associate recruitment, coordinate the MBG–Merck Biologics & Vaccines Symposium end-to-end, and manage corporate sponsorships

Director of Community Engagement, MIT Biotech Group (2024–2026)

Co-launched a \$5K scholarship with Harvard Biotech Club, now in its third year; organized equity-focused fireside chats connecting MBG members with MIT faculty and industry leaders

Projects Team Lead, Global Health Alliance (Fall 2024 – present)

Spearheaded two international partnerships producing women's health education materials – patient education workshops on maternal and child health for Village Health Works in Burundi, and a girls' education and empowerment curriculum with BIRD-Liberia; organized medical-debt forgiveness fundraisers erasing >\$100K at \$1 per \$100 forgiven

Camp Kesem Counselor (2024–present)

Fundraised >\$500 annually to support two free weeks of sleepaway camp for 200+ Boston-area children affected by a parent's cancer diagnosis; completed counselor training and provided mentorship and emotional support through week-long programming as part of MIT's student-run chapter

8.02 (Physics II) TA (Spring 2024)

Supported a flipped-classroom introductory E&M course by attending lectures, fielding student questions, and leading weekly Friday problem-solving sessions

20.A06 (Biological Engineering) TA (Fall 2024)

Improved the Biobots curriculum within MIT's freshman advising seminar, teaching biofabrication, 3D design, circuits, and neuronal cell and tissue culture

HONORS & AWARDS

Merck Prize, MIT Department of Biology (2025)

Awarded for outstanding research and academic performance in biophysical or bioinformatics sciences

Morais & Rosenblum UROP Award, MIT EECS (2025)

Departmental recognition for undergraduate research excellence

Platform Talk, American Society of Human Genetics Annual Meeting (2025)

Selected from top-scoring abstract submissions for a 10-minute oral presentation, <8% accepted

HKN Member, IEEE Eta Kappa Nu (2025)

Invitation-only honor society for EECS students; juniors in the top 25% of their class are eligible

EXPERIENCE

Summer Research Intern – Harvard Medical School Dept. of Biomedical Informatics (Summer 2026)

Supervisor: Prof. Marinka Zitnik

- Designed a modality-sensitivity benchmark for single-cell multimodal LLMs with a biologically inert baseline, gene order/identity scrambles, and graded marker-gene knockouts ($k = 1-25$) with expression-matched controls to isolate true biological signal from spurious noise
- Scaled the benchmark to a 300k-cell, 35-cell-type atlas via a sharded SLURM pipeline which benchmarks 5 model backends across 2 families (Cell2Sentence, CellWhisperer) using a two-tier, ontology-aware scoring scheme; manuscript for ICLR submission in progress

Undergraduate Researcher – MIT CSAIL (Fall 2023 – present)

Supervisor: Prof. Manolis Kellis

Evolutionary Comparative Genomics [*Winner of Merck Prize '25, EECS Research Award '25*]

- Developed ANCoRA, a damage-aware empirical Bayes algorithm for reference-guided diploid genome assembly, and applied it to build personalized genomes for 34 haplotypes across 17 individuals (5 modern humans, 4 archaic hominins, 8 great apes)
- Applied transformer-based deep learning to predict chromatin accessibility across 100 epigenomic contexts, defining 11k+ lineage-specific cis-regulatory elements (linCREs), which are largely undetected by prior divergence-based methods and enriched in neurodevelopmental pathways and disease loci
- Designed the variant-prioritization scheme that nominated a hominin-gained enhancer of GNB5 (linked to neurodevelopmental disorder), later validated by collaborators via luciferase reporter assays and reciprocal allele-swap experiments; manuscript submitted to *Nature Genetics*
- Presented these findings as an oral platform talk at the 2025 American Society of Human Genetics Annual Meeting, selected from the top 8% of submitted abstracts

Single-Cell Transcriptomics

- Applied the lab's Cell-Projected Phenotypes (CPP) framework to compute cell-level Parkinson's disease pathology scores and identify donors whose transcriptional states concord with case/control labels, addressing disease heterogeneity and enabling mechanistically grounded biomarker analysis

Course Research Project – MIT Deep Learning for Biology (Spring 2026)

- Independently designed and developed a multimodal model for generating cell-state-conditioned chromatin accessibility tracks in the context of Alzheimer's Disease with VAE embeddings for cell state and a discrete diffusion module

Computational Immunomics Intern – Infinitopes (Summer 2025) | Oxford, UK

Supervisor: Dr. Georges Bedran

- Built an end-to-end computational pipeline for generating, co-folding, and analyzing TCR-pMHC complex structures using Boltz-2, a state-of-the-art deep learning sequence-to-structure model
- Designed and executed five benchmarking studies to assess Boltz-2 performance in antigen recognition, structural accuracy, and immune specificity, identifying strengths and limitations of deep learning-based structure prediction for ranking tumor antigen immunogenicity

Genetic Medicines Scholar – Eli Lilly (January 2024) | Seaport, MA

- Pitched a novel genetic therapy for ALS – spanning target identification, drug delivery design, preclinical and clinical study planning, and regulatory approval strategy – as one of 12 MIT undergraduates selected for the inaugural cohort

Undergraduate Researcher – MIT Biological Engineering (June 2024 – December 2024)

Supervisor: Prof. Roger Kamm

- Engineered a neuronal tissue mold in CAD to prevent hypoxic necrotization by incorporating intentional void space for vascular ingrowth, addressing a key bottleneck in brain organoid viability
- Conducted on- and off-chip gelation trials to optimize gelatin-dextran hydrogel composition for synthetic vascular scaffolds in brain organoid models

Computational Neuroscience Researcher – UT Southwestern Biomedical Informatics (Summer 2023)

Supervisor: Prof. Wenhao Zhang

- Simulated Bayesian causal inference for multisensory cue combination in Python via importance sampling and Monte Carlo methods, modeling how the brain integrates visual and auditory stimuli into unified spatial percepts
- Implemented Langevin sampling for Bayesian causal inference, incorporating normalization constants and Gaussian noise to improve posterior estimation

Summer Intern – UT Southwestern Urology (Summer 2022)

Supervisor: Prof. Douglas Strand

- Selected for the UT Southwestern STARS Summer Research Program (< 5% accepted)
- Constructed a cellular anatomy for the female urethra and characterized two previously unknown distinct subtypes of fibroblasts in the female urethra with single-cell RNAseq data
- Performed in situ hybridization with RNAscope to validate the spatial separation of the fibroblast subtypes
- Developed proficiency with key single-cell analysis tools (Seurat, Harmony) for quality control and differential gene expression
- Dissected lower urinary tracts from human donors and sutured tissue in organ bath experiments

SKILLS

Computational & Programming: Python, PyTorch, R, Go, bash, C++, cluster computing, ML on AWS, Git/version control

Machine Learning & Genomics: transformer architectures, diffusion models, VAEs, single-cell analysis (Seurat, Harmony), sequence-to-function modeling

Experimental & Design: neuronal cell culture, brain organoids, biofabrication, tissue engineering, CAD (SolidWorks, Fusion 360), immunohistochemistry, western blot, qPCR, CRISPR, protein purification